

**YEAR 9 SCIENCE
BIOLOGICAL SCIENCES
TEST - BODY SYSTEMS**

NAME: SOLUTIONS

CLASS: _____

Mark: 48

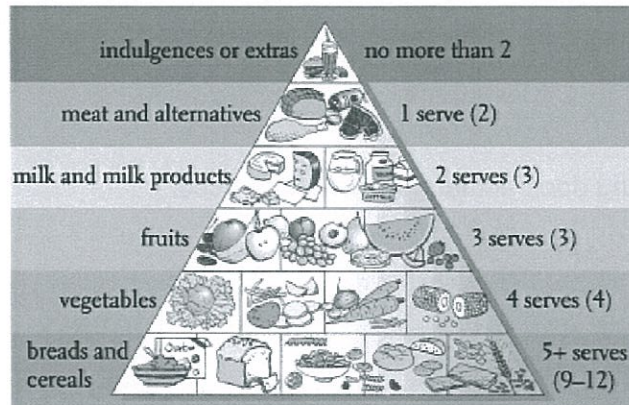
Multiple Choice Write the answer to each question in the appropriate box at right.

1. Which of the following lists contain only examples of **mechanical digestion**?
 - (a) Chewing in the mouth and the churning by stomach muscles.
 - (b) Chewing in the mouth, protease break down of proteins.
 - (c) Churning by the stomach muscles and protease break down of proteins.
 - (d) Protease break down of proteins is the only example.
2. Where is the digested food absorbed into your blood?
 - (a) Mouth.
 - (b) Large intestine.
 - (c) Small intestine.
 - (d) Stomach.
3. What are the main functions of the stomach?
 - (a) To store food so that you can use it when you are hungry.
 - (b) To break food down into a thick soupy liquid and to begin protein digestion.
 - (c) To make enzymes to digest fats, proteins and carbohydrates.
 - (d) To digest all the food you eat so it can be absorbed in the small intestine.
4. The **pancreas** can be described as which of the following?
 - (a) An organ that produces enzymes.
 - (b) A site of chemical digestion.
 - (c) A site of mechanical digestion.
 - (d) An organ of excretion.
5. Absorption of water in the digestive system occurs in which of the following organs?
 - (a) The stomach.
 - (b) Large intestine.
 - (c) Liver.
 - (d) Anus.

QUESTION	ANSWER
1	A
2	C
3	B
4	A
5	B
6	C
7	C
8	D
9	D
10	D
11	A
12	A
13	C
14	D
15	D
16	A
17	C
18	B
19	A
20	C
21	A
22	B
23	C

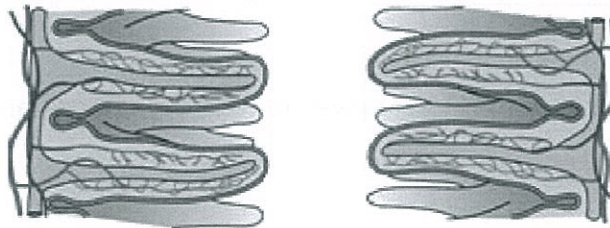
6. When urine leaves the bladder it:
- (a) is taken to the kidney.
 - (b) moves through the ureter.
 - (c) moves through the urethra.
 - (d) is taken to the liver.
7. What is the purpose of digesting your food?
- (a) So that the muscles in your oesophagus can squeeze the pieces down to your stomach.
 - (b) So that you can break it into pieces small enough to swallow.
 - (c) To break it into molecules small enough to be absorbed into your blood.
 - (d) The food keeps your stomach and intestines alive by giving them energy.
8. How do the ***circulatory*** and ***excretory systems*** co-operate to keep you alive?
- (a) The circulatory system, via the blood, delivers food to the kidneys, which then deliver wastes to the blood.
 - (b) The circulatory system, via the blood, delivers digested foods to the kidneys to keep them alive.
 - (c) The circulatory system, via the blood, delivers wastes to the kidneys, which then pass urine to the blood to be carried to the bladder.
 - (d) The circulatory system, via the blood, delivers wastes to the kidneys that they then excrete from the blood by forming urine.
9. A number of nutrients provide our bodies with energy. The main source of energy is:
- (a) vitamins.
 - (b) protein.
 - (c) lipids.
 - (d) carbohydrates.
10. Two nutrient groups are required in minute (very small) amounts for healthy growth. They are:
- (a) vitamins and carbohydrates.
 - (b) minerals and proteins.
 - (c) carbohydrates and proteins.
 - (d) vitamins and minerals.
11. Which of the following lists some functions of the blood?
- (a) Transport of oxygen, food, wastes and heat.
 - (b) Carrying food, breathing, fighting infection and warming the body.
 - (c) Transporting oxygen and food, removing wastes and digesting food.
 - (d) Fighting infection, carrying food, breathing and digestion.
12. What do we call the heart chambers which pump blood out of the heart into the arteries?
- (a) Left and right ventricles.
 - (b) Left ventricle and left atrium.
 - (c) Right atrium and right ventricle.
 - (d) Left atrium and right atrium.

13. According to the CSIRO's recommendations for healthy eating, people need to consume 3 serves a day of:



- (a) meat and alternatives.
(b) bread and cereals.
(c) fruit.
(d) indulgences or extras.
14. What is the name of the group that includes the nutrients starch and sugar?
- (a) Vitamins.
(b) Lipids.
(c) Fats.
(d) Carbohydrates.
15. The lungs are ventilated by muscular movements of the rib cage and the ____.
- (a) trachea
(b) ribs
(c) alveoli
(d) diaphragm
16. ____ carry blood away from the heart.
- (a) Arteries
(b) Bronchioles
(c) Veins
(d) Venules
17. The millions of tiny air sacs in your lungs are called ____.
- (a) capillaries
(b) bronchioles
(c) alveoli
(d) bronchi
18. The lower end of the trachea divides into the left and right ____.
- (a) oesophagus
(b) bronchi
(c) alveoli
(d) trachea

19. The smallest blood vessels in your body are called ____.
- (a) capillaries
 - (b) venules
 - (c) arteries
 - (d) arterioles
20. In what part of the lung does actual gas exchange occur?
- (a) The bronchiole.
 - (b) The trachea.
 - (c) The alveolus.
 - (d) The bronchus.
21. Backflow of blood back into the heart is prevented by ____, which allow blood to flow only one way.
- (a) valves
 - (b) arteries
 - (c) atria
 - (d) veins
22. A cross-section of the **small intestine** shows that it is lined with tiny finger-like projections. What is the name of these projections?

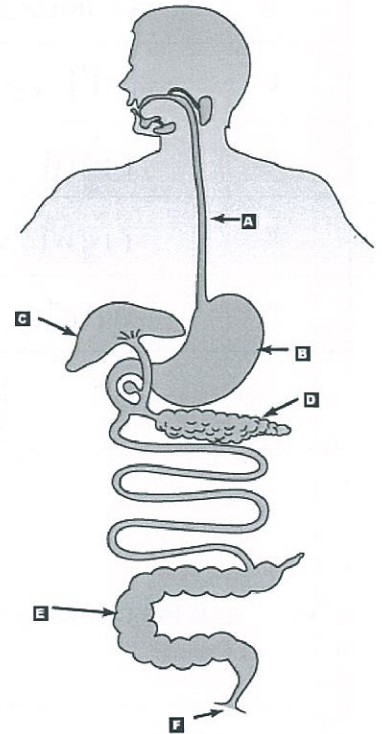


- (a) Nephrons.
 - (b) Villi.
 - (c) Filaments.
 - (d) Capillaries.
23. The chamber of the heart which has the thickest wall is the ____.
- (a) right atrium
 - (b) left atrium
 - (c) left ventricle
 - (d) right ventricle

Short Answer**Write the answer to the questions in the spaces provided.**

1. Write the name of the parts shown in the following diagram of the digestive system. (6 marks)

PART	NAME
A	oesophagus (gullet)
B	stomach
C	liver
D	pancreas
E	large intestine
F	anus



* Be lenient with spelling. [1 mark each]

2. There are three enzymes that target the three major food groups. Complete the following table. (6 marks)

NAME OF ENZYME	TARGET FOOD GROUP
amylase	carbohydrate
protease	protein
lipase	fats + oils

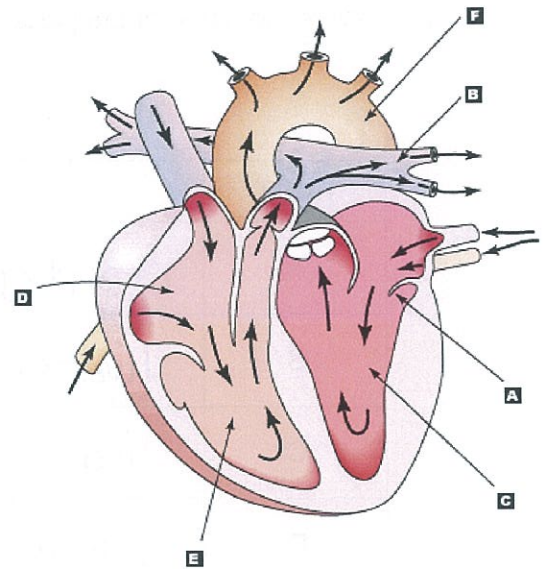
[1 mark each]

3. Label the parts of the heart marked A to F.

(6 marks)

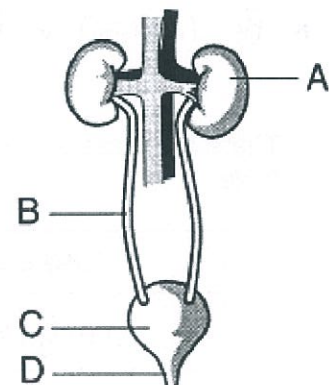
A	valve
B	pulmonary artery
C	left ventricle
D	right atrium
E	right ventricle
F	aorta

[1 mark each]



4. Use the following diagram of the human excretory system to help complete the table below. (4 marks)

LETTER	NAME OF STRUCTURE
A	kidney
B	ureter
C	bladder
D	urethra



5. (a) Write a word equation for **respiration**.

(1 mark)



* Be generous!

(b) What are the **waste products** of respiration?

(2 marks)

CO_2 , H_2O , urea
 1 mark 1 mark